

SEOYOON YUM

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

2022 – Present

B.S. in Computer Science

Selected coursework: Machine Learning, Deep Learning, Operating Systems, Computer Architecture, Systems Programming, Algorithms.

RESEARCH EXPERIENCE

Research Intern, SEAL, KAIST

Sep. 2026 – Present

Advisor: Prof. Sehoon Kim

- Investigating optimization techniques for looped transformers.

Research Intern, CHEI Lab, UC San Diego

Jun. 2026 – Aug. 2026

Advisor: Prof. Falko Kuester

- Built a zero-shot anomaly-detection pipeline using the frozen Toto 2.0 time-series foundation model on production Ceph and DigitalPath telemetry.
- Evaluated detection of proxy anomalies and alert-volume tradeoffs; showed that forecast residuals did not reliably predict genuine failures.

Independent Researcher, PhaseGate

Jun. 2026 – Present

Self-initiated and led; later advised by Prof. Sehoon Kim, KAIST

- Built a phase-aware CPU retrieval scheduler for on-device LLMs, achieving $2.0\times$ the throughput of the best feasible fixed policy under the same latency constraint.
- Identified asymmetric CPU–GPU memory contention: four concurrent HNSW searches increased p95 decode latency by 60–61%, versus 5.7–6.9% for prefill, across two Apple M4 systems.

Research Intern, SGVR Lab, KAIST

Jan. 2026

Advisor: Prof. Sung-Eui Yoon

- Evaluated Octo-Small vision-language-action policies on SIMPLER under 11 meaning-preserving perturbations; average success fell from 60% to 25%, with structural edits causing the largest drop.

Research Intern, CJ AI Center

Summer 2023

- Built a real-time brain–computer interface mapping EEG signals to robotic-arm commands using PyTorch.

SELECTED WORK

- [1] **S. Yum** and S. Kim, “PhaseGate: Phase-Aware CPU Retrieval Scheduling for On-Device LLMs on Unified Memory.” Manuscript under review at a NeurIPS 2026 ODI workshop.
- [2] **S. Yum**, “Zero-Shot Time-Series Foundation Models for Infrastructure Anomaly Detection.” Technical report, 2026.
- [3] **S. Yum**, “Stress-Testing Semantic Robustness in Vision-Language-Action Policies.” Technical report, 2026. doi:10.5281/zenodo.20777663.

LEADERSHIP

Captain, KAIST American Football Club

Sep. 2025 – Present

- Lead team practices, game strategy, and intra-university competitions.

Squad Leader and English Interpreter

Jan. 2024 – Jul. 2025

Republic of Korea Army

- Led eight soldiers and interpreted between Republic of Korea and U.S. military personnel.

TECHNICAL SKILLS

Code and tools

Python, C, Bash, Git, Linux

ML and systems

PyTorch, MLX/MLX-LM, FAISS/HNSW, Prometheus, Grafana, Ceph

Languages

Korean (native), English (professional; TOEFL 110)